INTRODUCTION OF COMPANY

1. Company building having three floors like ground floor, 1st floor and second floor.
2. Total 53 devices out of which 46 mobile devices and 07 are fixed.
3. Total No of employees are 21.
4. The workforce includes:

- 4 Equipment handlers
- 1 Equipment manager
- 1 Head chef
- 10 Chefs
- 1 Catering manager
- 3 Office workers
- CEO (owner)

5. The table below outlines the employees and their associated devices.

	Office Worker	Equipment Handler	Equipment Manager	Transient Staff	Catering Manage	Head Chef	Chefs	CEO	Total devices
Desktop Computer	3		1		1	1		1	7
Laptop Computer							3		3
Tablet		4	1		1	1	3	1	11
Mobile Phone				30	1			1	32

6. VIP Food app description.
   The app offers a range of specialized functionalities, including:

- Equipment management
- Kitchen management
- Event management
- Administration
- Day-to-day tasks

7. Different employees access to the app as per their roles, such as define below.

   Equipment Handlers Role (EquipHandlersRole):
   Tailored for equipment handlers, this role provides access to essential equipment management functionalities within the VIP Foods App. It enables efficient tracking and handling of culinary tools and resources.

Equipment Manager Role (EquipManagerRole):
Designed for equipment managers, this role offers advanced capabilities to oversee and manage the full spectrum of equipment functionalities. It ensures optimal utilization and maintenance of culinary resources.
Chefs Role (ChefsRole):
Crafted for chefs, this role grants access to food preparation and kitchen management functionalities. Chefs can leverage the app to streamline culinary processes and enhance collaboration within the kitchen.
Head Chef Role (HeadChefRole):
Specifically designed for head chefs, this role provides elevated access to overall kitchen management functionalities. It empowers head chefs with tools to efficiently coordinate and lead culinary operations.
Catering Manager Role (CateringManagerRole):
Tailored for catering managers, this role facilitates planning and managing catering events. It includes features for event coordination, menu planning, and logistics management to ensure seamless catering services.
Office Workers Role (OfficeWorkersRole):
Customized for office workers, this role provides access to office-related activities and data management functionalities within the VIP Foods App. It enhances communication and collaboration between the culinary and administrative teams.
CEO (Owner) Role (CEORole):
Specifically designed for the CEO or owner, this role grants unrestricted access to all functionalities on the VIP Foods App. It enables strategic oversight, and decision-making, and ensures comprehensive control over the platform.
Temporary Role (TempRole):
Assigned to temporary workers, this role offers limited access to permissions required for the day's tasks. The permissions are automatically disabled after a predefined period, ensuring temporary and secure access tailored to the user's immediate needs.

STAGE 01

In stage 01 we have 06 tasks to do.

Task01 => Understand the Requirements
Task02=> Understand the building structure and planning secure network installation.
Task03=> Access control and Security Policies
Task04=> User roles and provide access
Task05=> Physical Security

Now we discuss on each task in details.

Task 01: Understand the Requirements.
From the introduction it is clear, we must provide cybersecurity to the company and company have 21 employees, some wired and some wireless internet devices, company also have their app. Now we must provide such a security level in which company employees gained access to the app securely as per their role access with physical security of devices and much more security aspect which we discuss in later stages.

Task 02: Understand the building structure and planning secure network installation.
The network architecture for the new building is guided by defense in depth principles to establish a comprehensive layered security approach. Network segmentation should be implemented to isolate critical systems and data, reducing the risk of potential attacks.

- Equipment handlers:
  Wired: 10.0.1.0/26
  Wireless: 10.0.1.128/26
- Equipment manager:
  Wired: 10.0.1.64/26
  Wireless: 10.0.1.192/26

First floor subnets:

- Chefs:
  Wired: 10.0.2.0/26
  Wireless: 10.0.2.128/26
- Head chef:
  Wired: 10.0.2.64/26
  Wireless: 10.0.2.192/26
- Catering manager:
  Wired: 10.0.2.128/26
  Wireless: 10.0.2.0/26

Second floor subnets:

- Office workers:
  Wired: 10.0.3.0/26
  Wireless: 10.0.3.128/26

- CEO (Owner):
  Wired: 10.0.3.64/26
  Wireless: 10.0.3.192/26
  Production subnet:
- Servers, printers, POS systems:
  Wired: 10.1.0.0/24
  Wireless: 10.1.0.128/25
  Management subnets:
- Routers, switches, firewalls:
  Wired: 10.1.1.0/24
  Wireless: 10.1.1.128/25
- Guest and transient worker subnet:
  Laptops, tablets, smartphones: 10.1.2.0/24
  Wireless: 10.1.2.128/25

Task03=> Access control and Security Policies
User access to specific subnets should be facilitated through precisely outlined policies overseen by the on-premises firewall.
install firewalls at the network perimeter to efficiently filter both incoming and outgoing traffic. Firewalls should be configured to analyze and permit legitimate data while proactively blocking potentially harmful content.
deploy intrusion detection/prevention systems (IDS/IPS) at the network perimeter. Utilize IDS/IPS systems to continuously monitor network activities, ensuring prompt detection of and response to any suspicious behavior or potential intrusion attempts.

Task04=> User roles and provide access
Azure Active Directory (Azure AD) implementation: An Azure AD tenant should be configured and user accounts should be created based on employee roles. It is recommended to use group-based access control with a twist—assign roles directly within the groups. By doing so, group members will automatically inherit the specified role, ensuring a seamless and efficient approach to resource access management according to user groups.

Task05=> Physical Security
This boring me to discuss but let start having dedicated IT Room for server router and switch with restriction of unauthorized access, ID Based Access CCTV Cameras locked bela bela.

STAGE 02

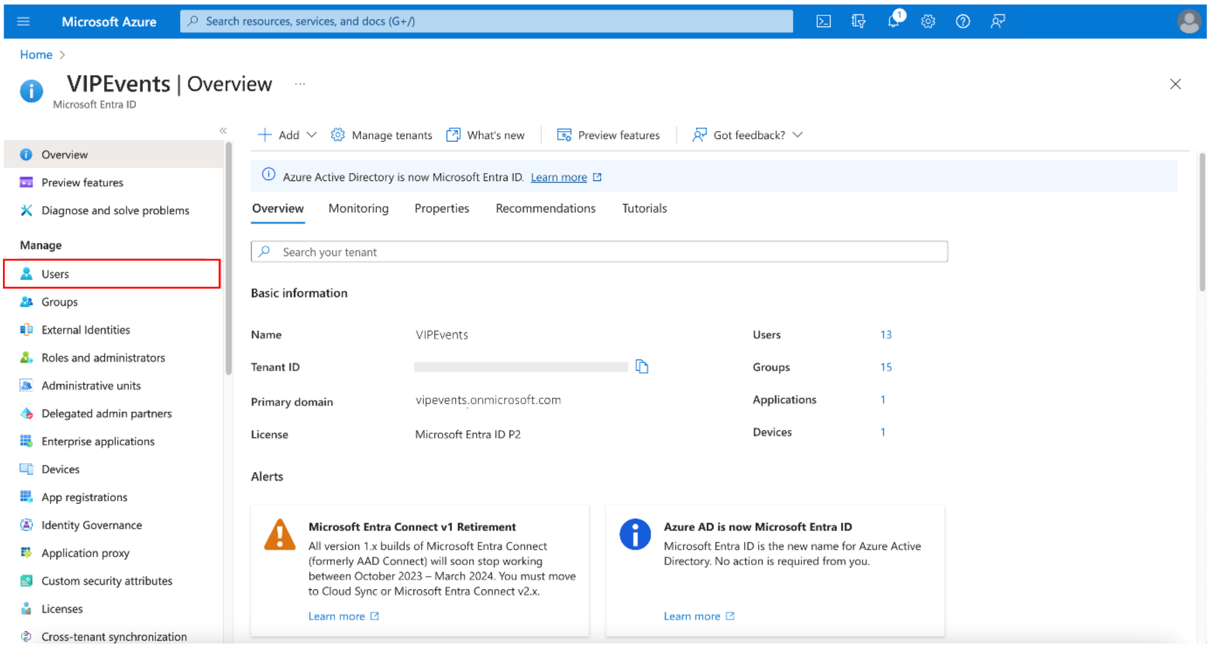
AZURE ACTIVE DIRECTORY SETUP

Task 01 => Azure AD Tenant Creation.

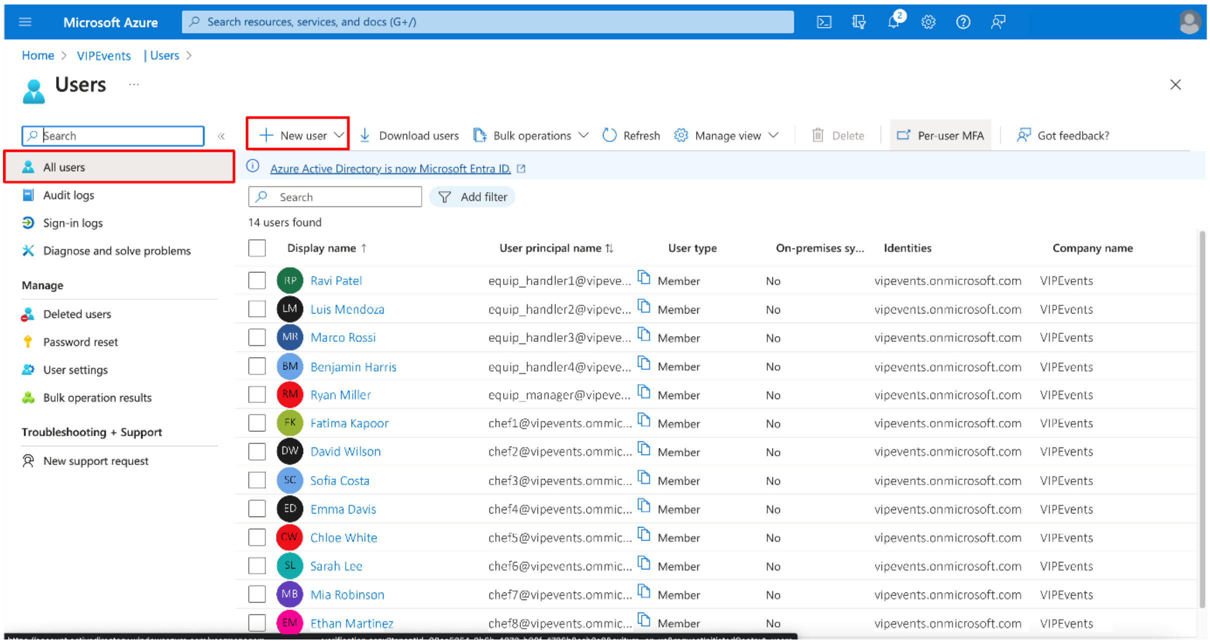
An Azure AD tenant should be created for VIP Events to ensure seamless integration with the organization's infrastructure

Task 02 => User Account Configuration

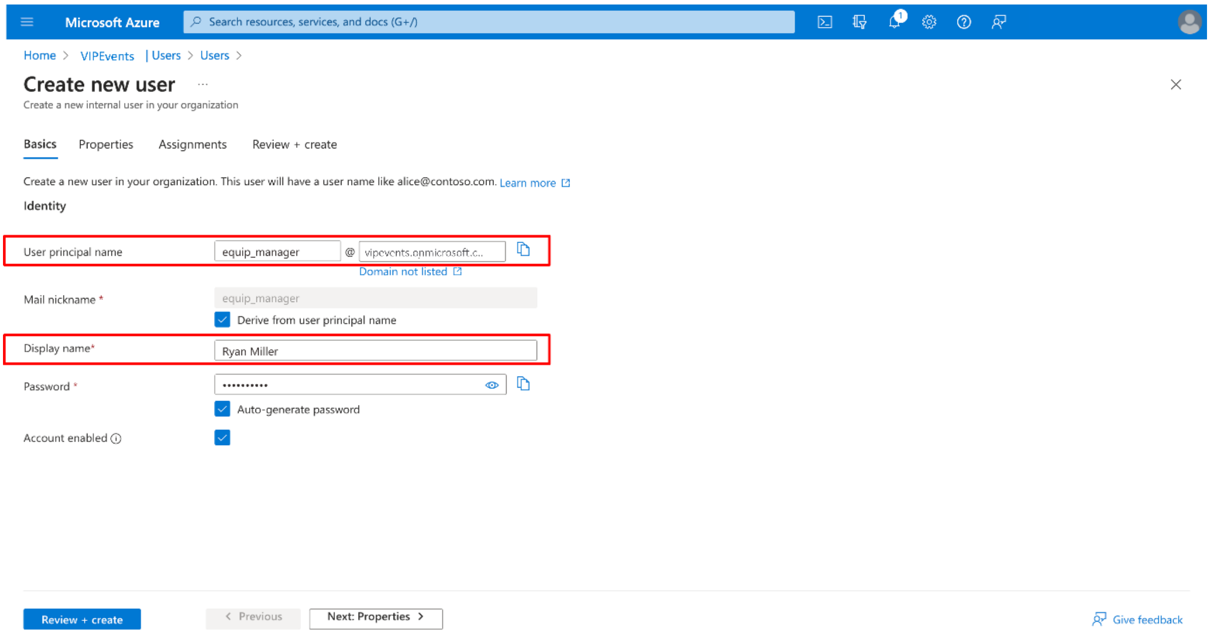
1. User Account Creation
   Access the **Microsoft Entra Admin Center** and sign in with privileges equivalent to a User Administrator (at minimum).



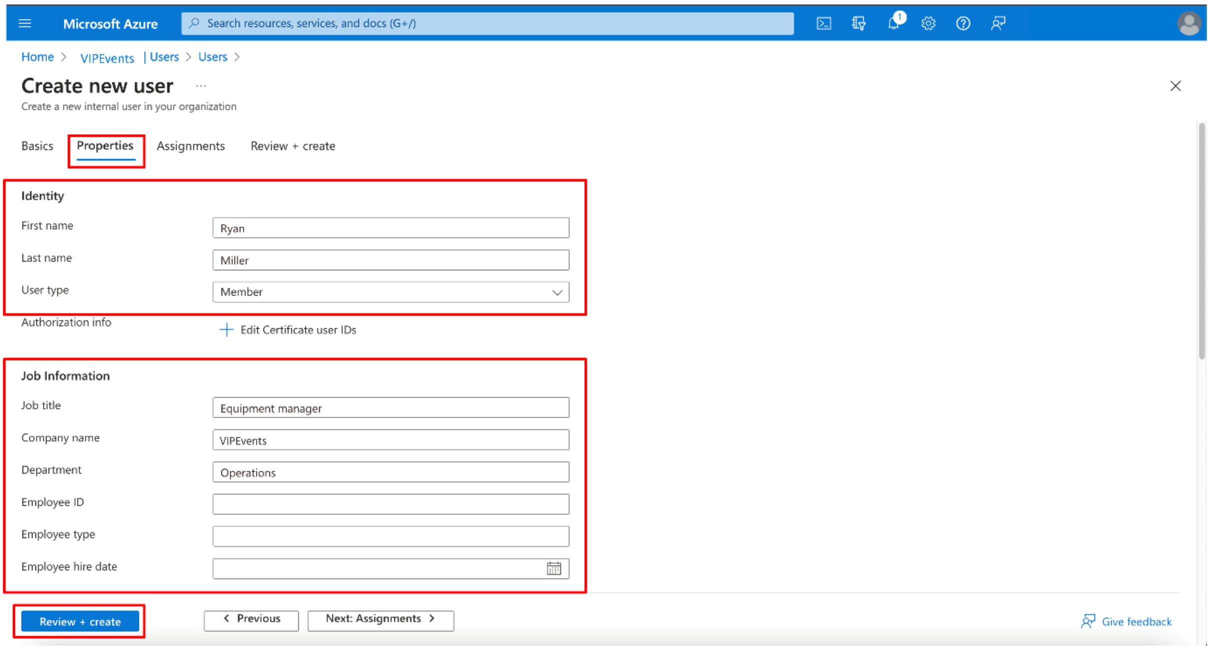
2. Navigate to **User Creation.**
3. Scroll down to **Manage**, select **Users** and then **All users**.



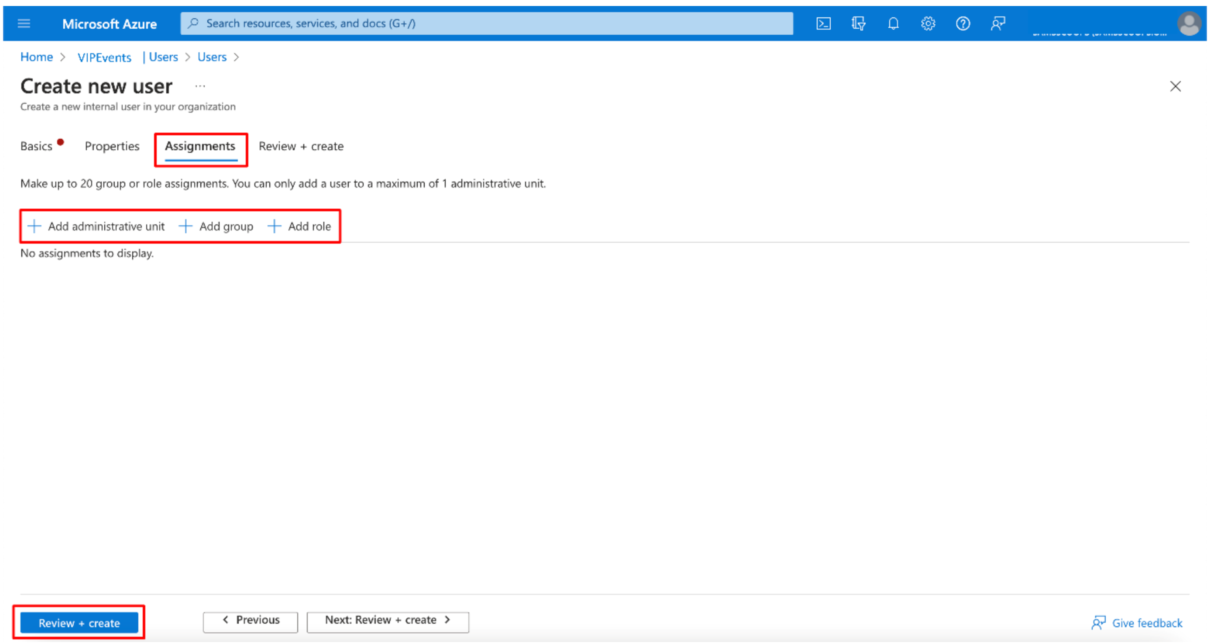
4. Create a new user by selecting +**New user** at the top of the pane and then select **Create new user.**
5. In the **Basics** tab, add user account details. In the **User principal name** field add the usernames provided in the table below (e.g. “equip_manager @vipevents.onmicrosoft.com”), adjusting the domain as needed.
6. In the **Display name** field, enter the user's name.



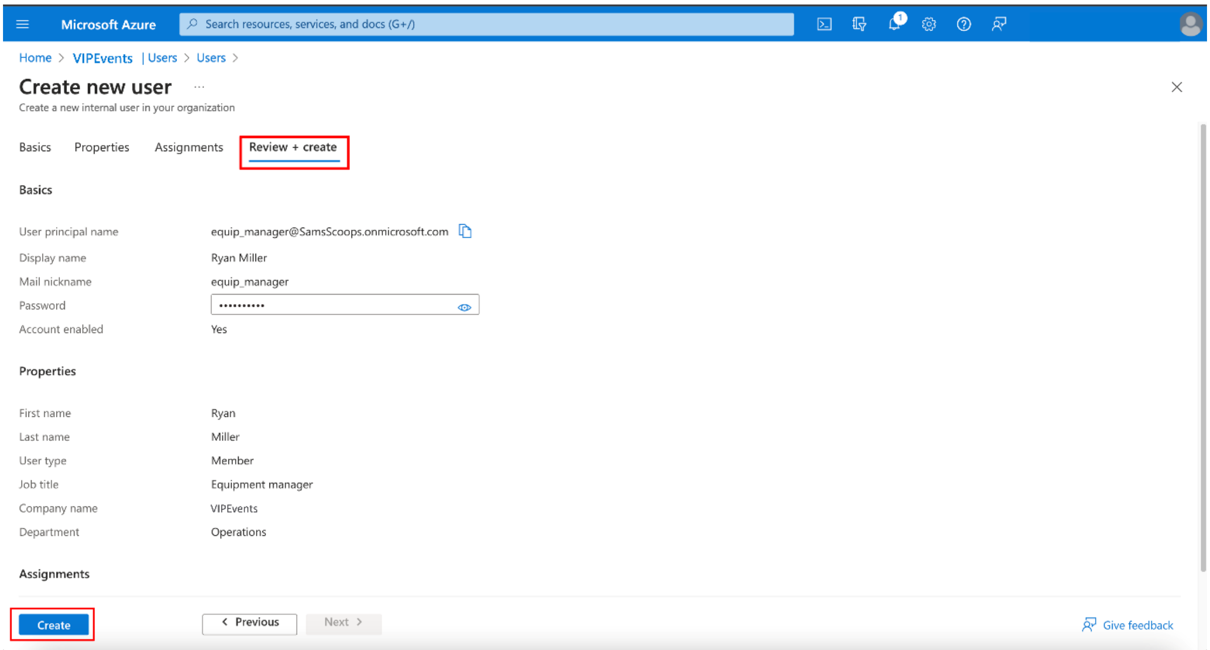
7. Under the Properties tab, add the **Job role** and **Department** provided in the table for each user.



8. Under the **Assignments** tab, enter the required details for the user under the **Groups and roles**, **Settings**, and **Job info** sections.



9. To finalize the user creation, select **Review+Create** and then **Create**.



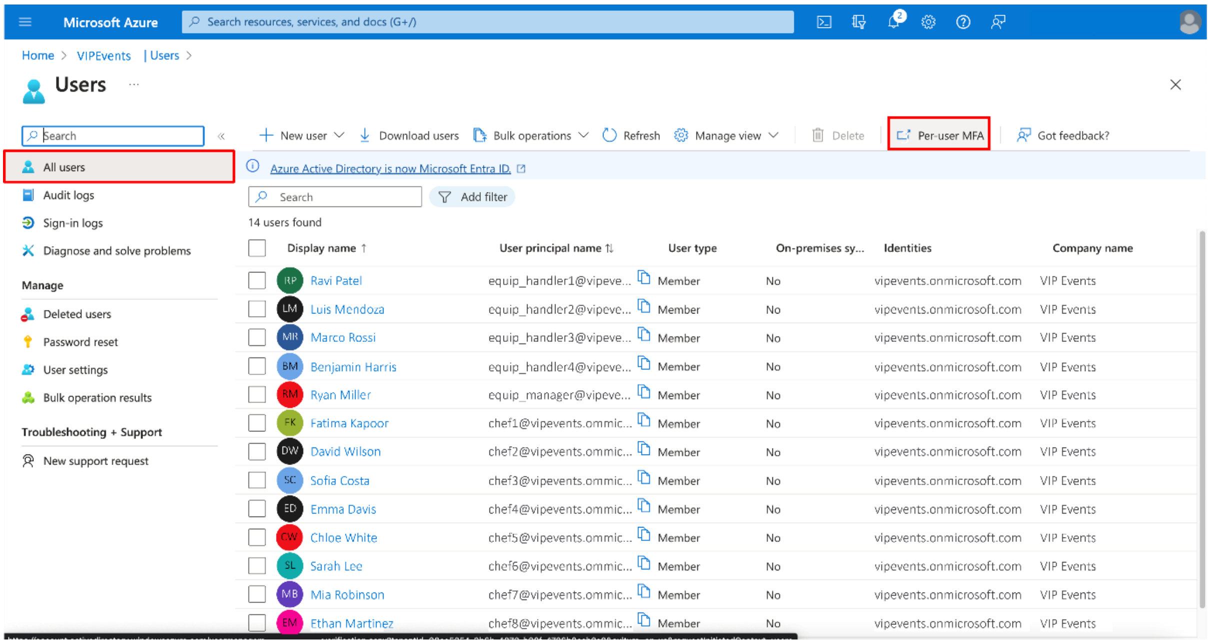
The table below provides user details for VIP Events employees.

Note: Usernames used here were randomly generated, you can use any user names.

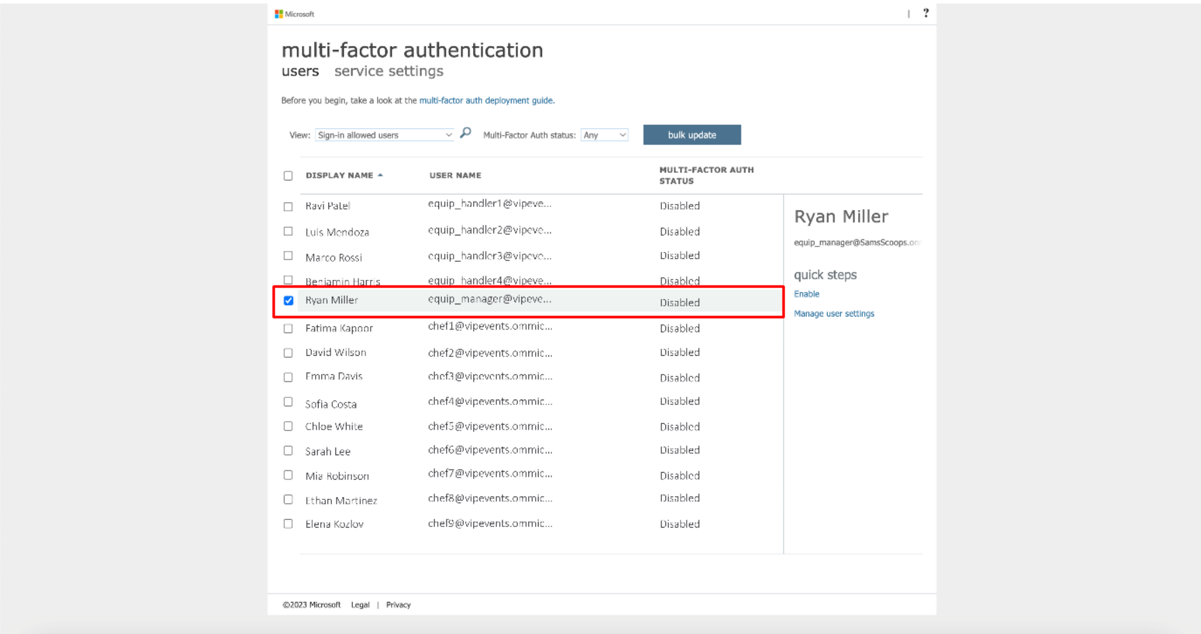
Username	Display name	Job role	Department
equip_handler1	Ravi Patel	Equipment handler	Operations
equip_handler2	Luis Mendoza	Equipment handler	Operations

Security should be enhanced by enabling multi-factor authentication for all user accounts. To do this follow the steps outlined below:

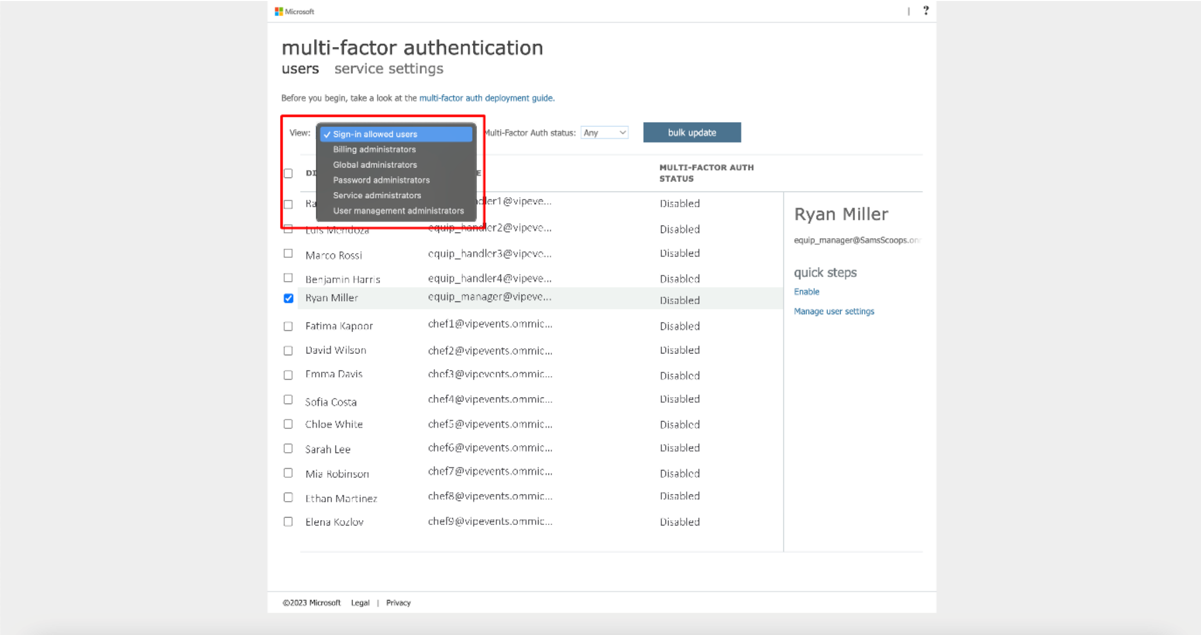
1. Navigate to the Microsoft Entra **multi-factor authentication** users page.



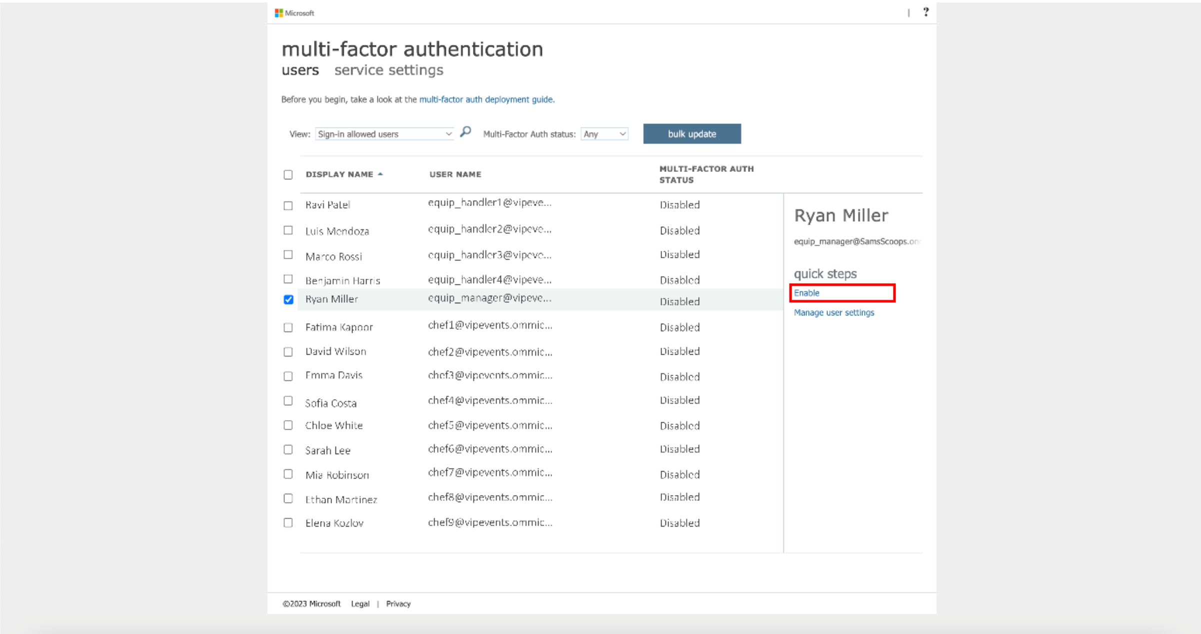
2. To check if MFA is enabled for a user, locate and select the user as demonstrated in the screenshot below.



3. Adjust the view to **User** if needed.



4. Select the user(s) by checking the box next to their names.
5. To enable multi-factor authentication, select **Enable** on the panel on the right of the screen, under **quick steps**.



6. Confirm your selection in the pop-up window that appears by selecting **enable multi-factor auth**.

Task03 => **Group-based access control implementation**

To implement a group-based access control system, users should be categorized into distinct groups based on their roles. The following are the identified user groups along with their respective members:

- Equipment handler group:

Members: equip_handler1, equip_handler2, equip_handler3, equip_handler4

- Equipment manager group:

Members: equip_manager

- Chefs group:

Members: chef1, chef2, chef3, chef4, chef5, chef6, chef7, chef8, chef9, chef10

- Head chef group:

Members: head_chef

- Catering manager group:

Members: catering_manager

- Office workers group:

Members: office_worker1, office_worker2, office_worker3

- CEO (Owner) group:

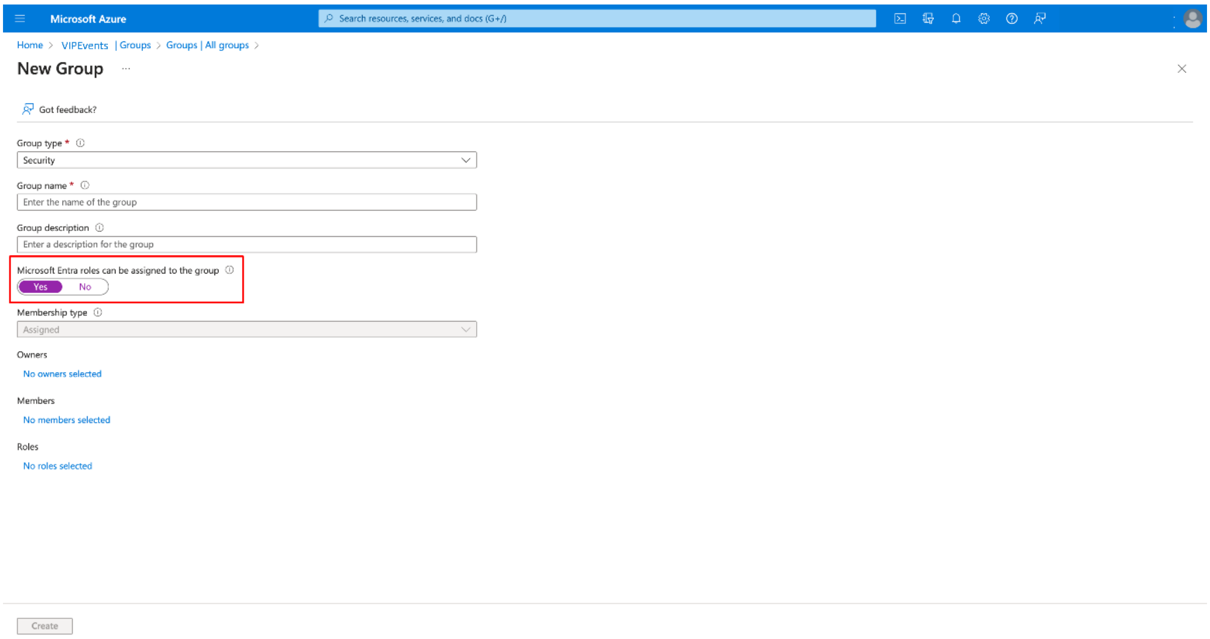
Members: ceo_owner

- Worker - External:

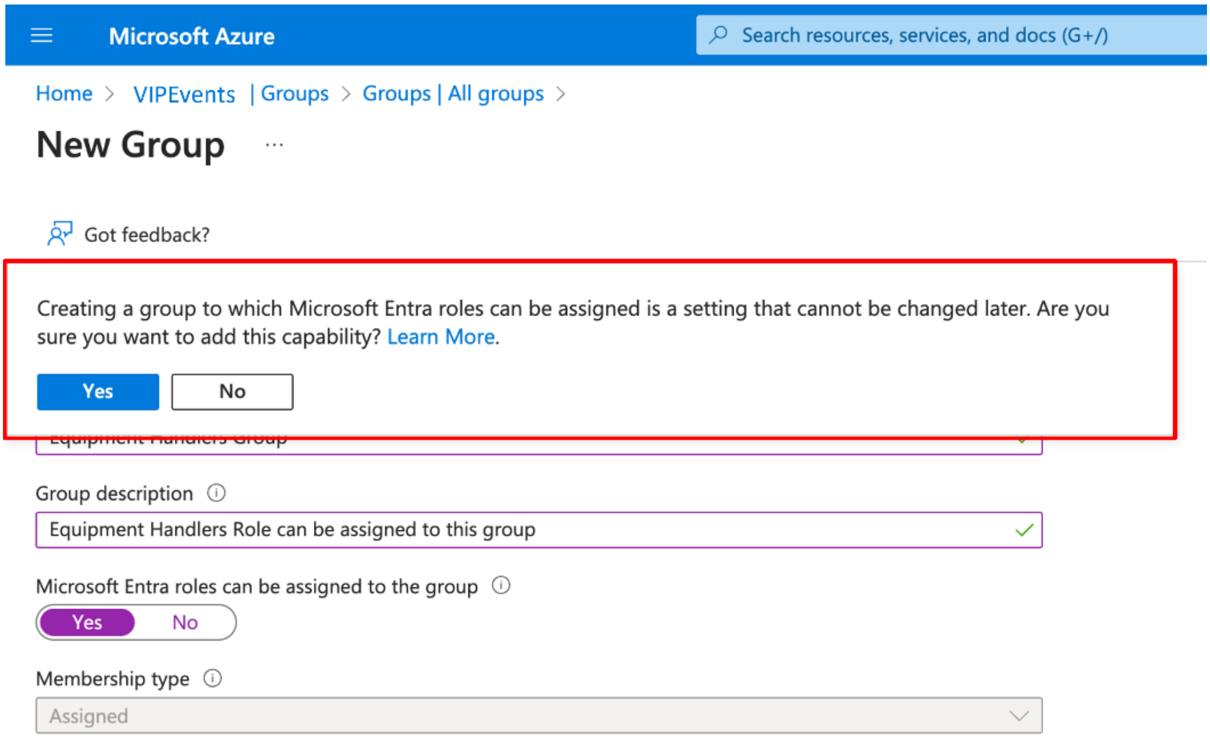
Members: user1_ext

Dedicated security groups should be established within Azure AD that align with the identified user roles. To do this, follow these steps:

1. Access the **Microsoft Entra Admin Center** as a Global Admin.
2. Navigate to **Group Creation**, scroll down to Identity, select **Groups** and then **All groups**.
3. To initiate a new group, select **New group**.
4. Configure the group details by adding the required group type, name, and description.
5. Enable role assignment for the group: Set **Microsoft Entra roles can be assigned to the group** to **Yes**.



6. Select members and owners for the group.
7. Assign roles to the group.
8. To confirm the group creation, select **Create**.
9. Acknowledge the irreversible nature of the role-assignable group setting when prompted.
10. Finalize the group creation by selecting **Yes**.



To map user groups to Azure roles, you should integrate individual user accounts into the access control framework by adding them to their corresponding security groups. This ensures a smooth and organized implementation of the group-based access control system.

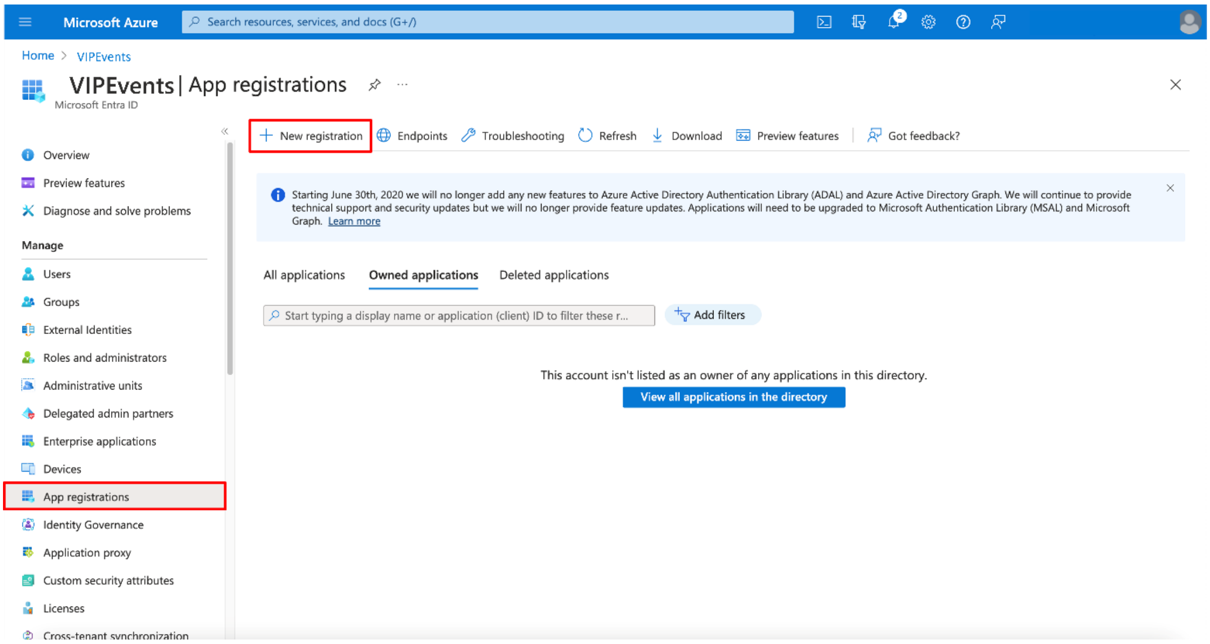
Stage 03 (Define Roles and Access)

This section describes the proposed Azure role and access configuration. These roles, tailored to the unique needs of VIP Events, are strategically assigned based on the user groups established earlier, creating a precise and secure access control mechanism.

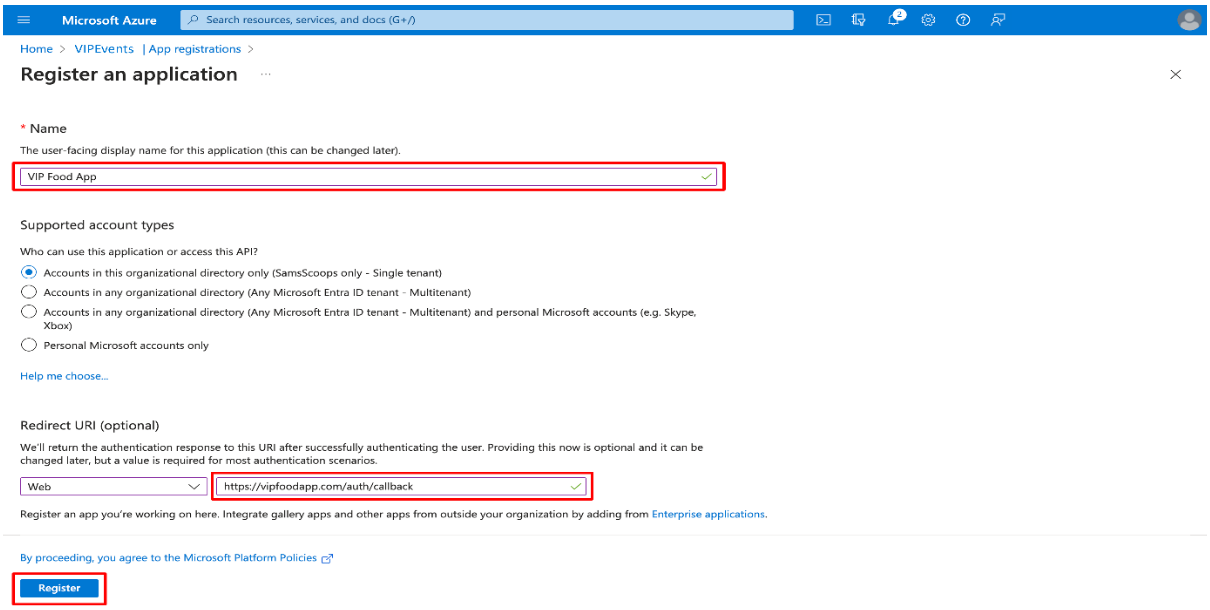
VIP Foods application integration (Task 1)

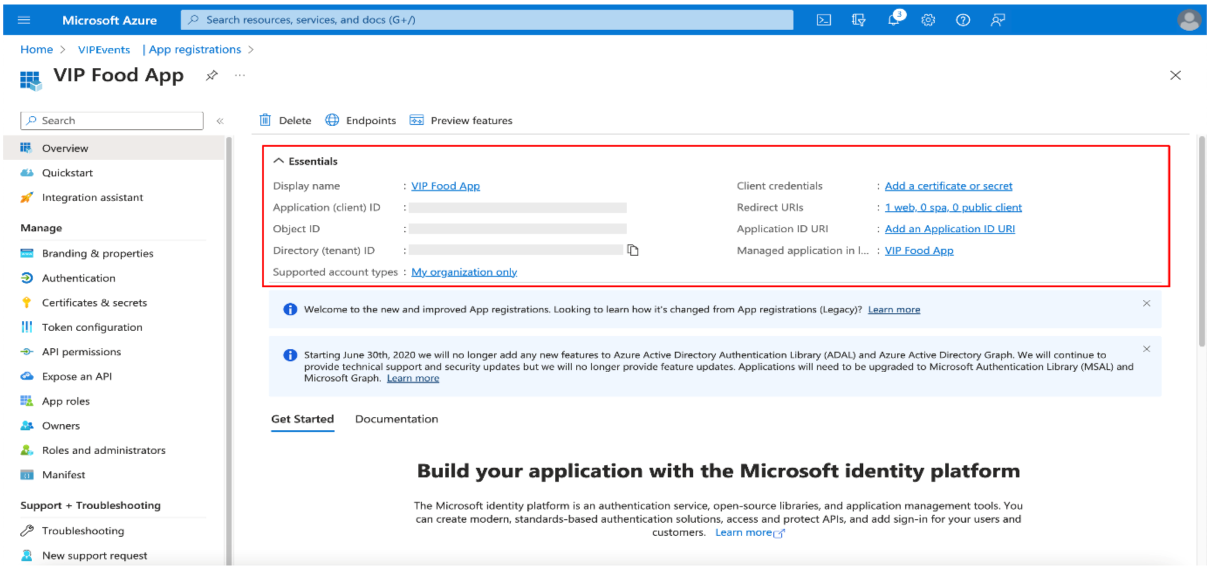
The first step that needs to be performed to allow for role-based access control is to create an application registration for the VIP Food app. The key steps involved are as follows:

1. Sign in to the **Microsoft Entra admin center**
2. Navigate to **Manage**, select **Applications** then **App registrations** and finally select **New registration**.

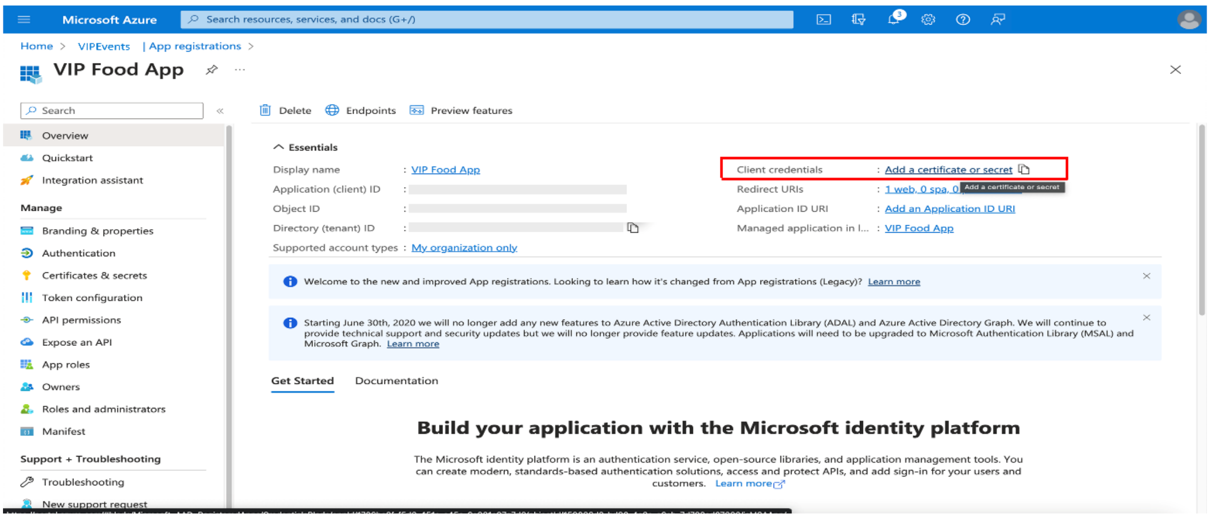


3. Enter a display name for the application, specify the sign-in audience, register the application, and note the Application ID (client).
4. Add a redirect URI for the VIP Food app.
5. And select **Register**.

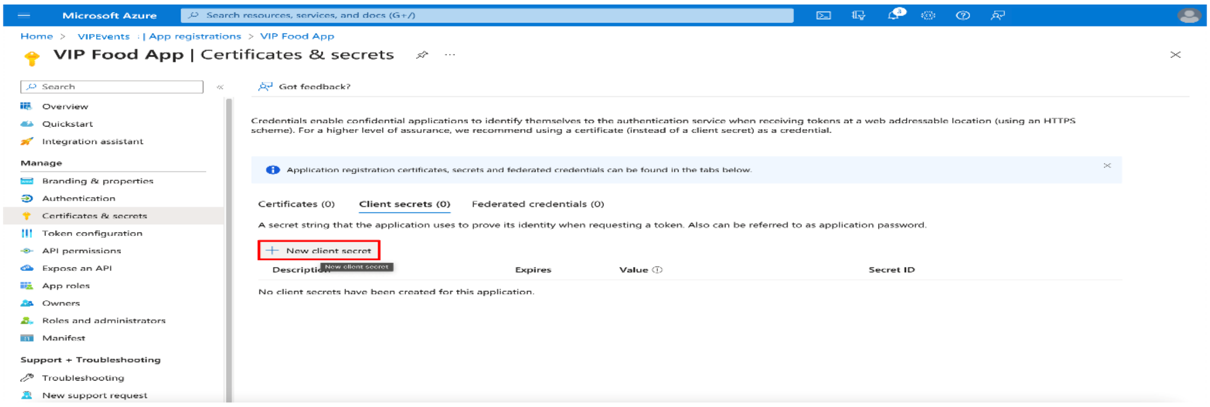


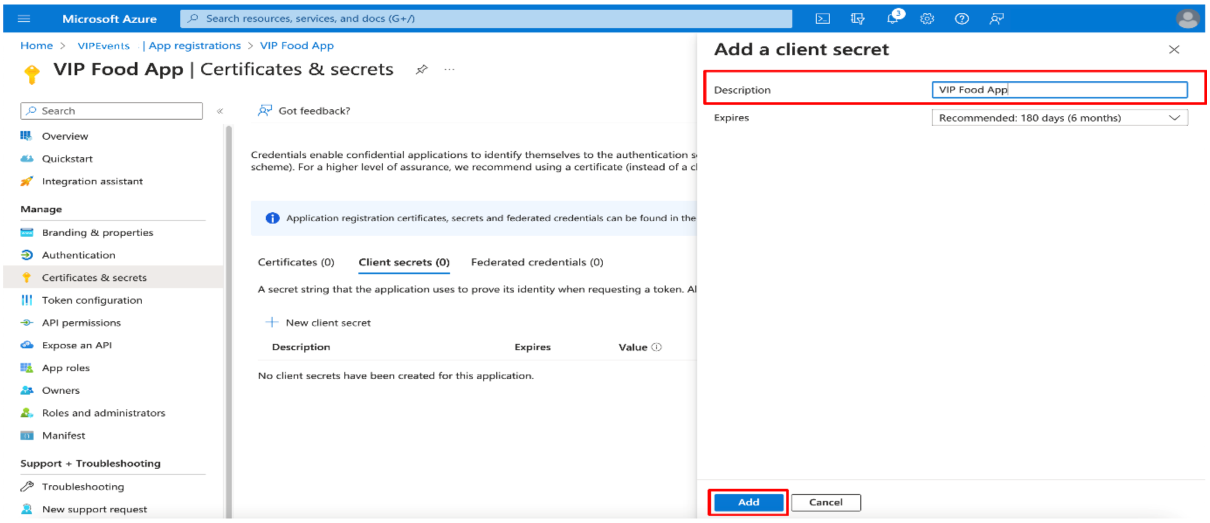


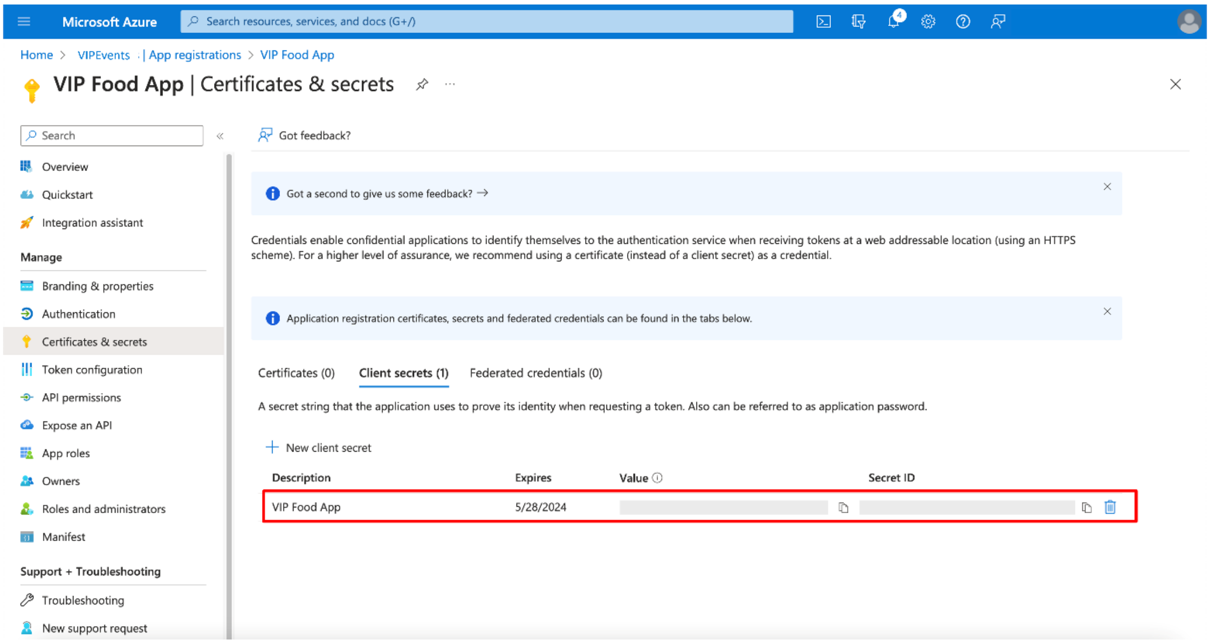
6. Add a client secret.



7. Select **+ New client secret**.







1. Document these details.

To facilitate clarity and future reference, the following details should be documented.

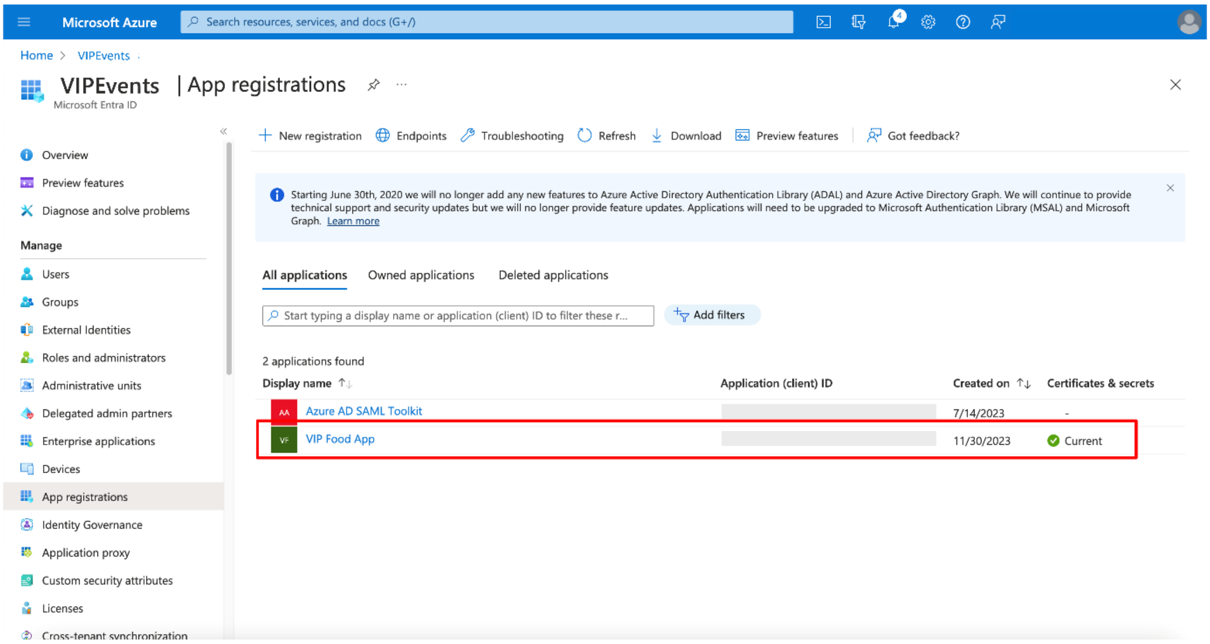
Registered app details:

- Name: VIP Food App
- Client ID: 17546bc3f-f5d3-451a-a45c-6a831a87a5d3
- Tenant ID: 98ce5354-3b6b-9735-b80f-4826b0ccb3a9
- Client secret
  - Description: VIP Food App
  - Expires: 5/28/2024
  - Value: Elb8Q~UmtQcbSXgQ_2dHnClaNOWOTMrP_.P8da69
  - Secret ID: 79e9994f-0197-4607-ad6d-b4fc4342baed
- Redirect URI: Web: https://{URL}/auth/callbackStage 4: AAD Connections

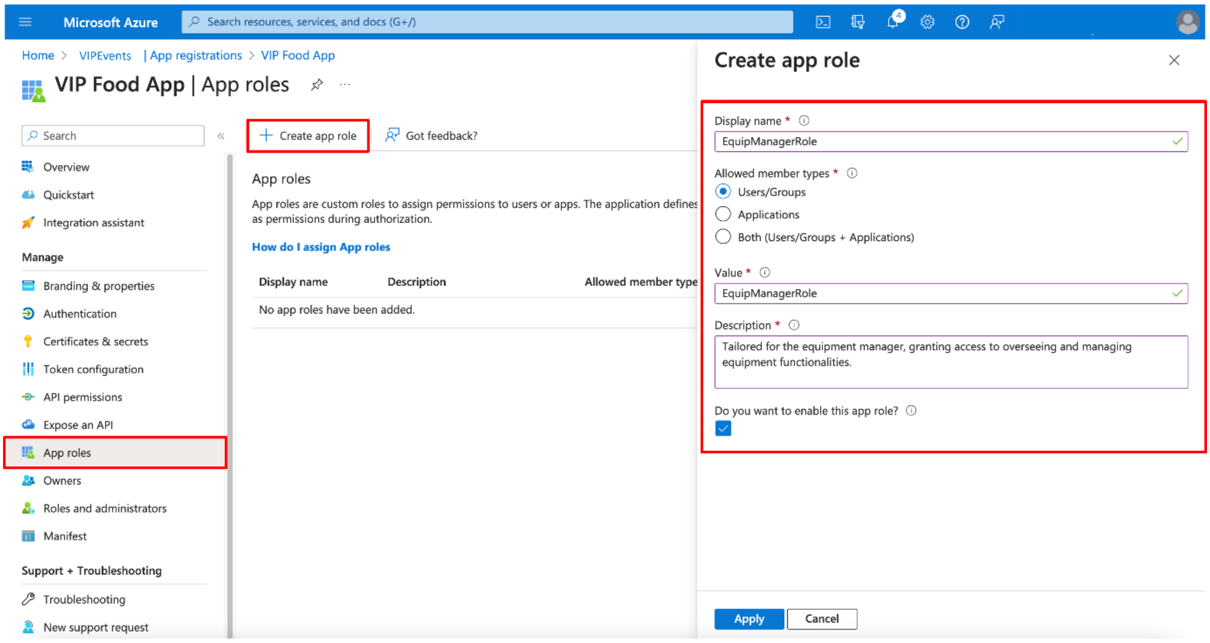
Custom Azure Roles (Task 2)

To align roles with user groups and facilitate controlled access, custom Azure roles are needed. The process of creating custom Azure roles involves the following steps:

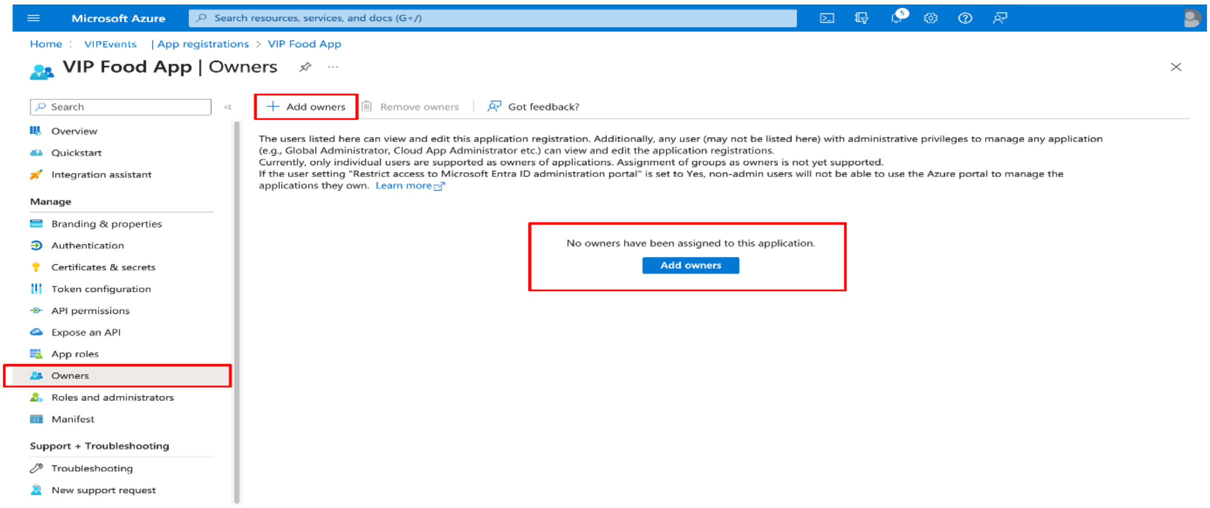
1. Sign in to the **Microsoft Entra admin center** as a Global Administrator.
2. Navigate to **Identity**, select **Applications,** and then **App registrations**. Select the relevant application here.

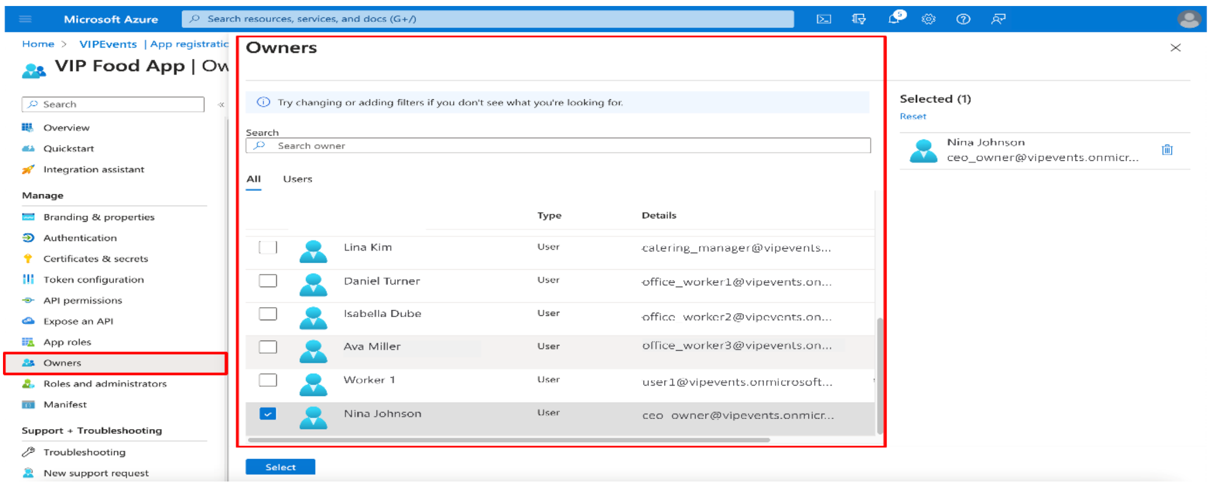


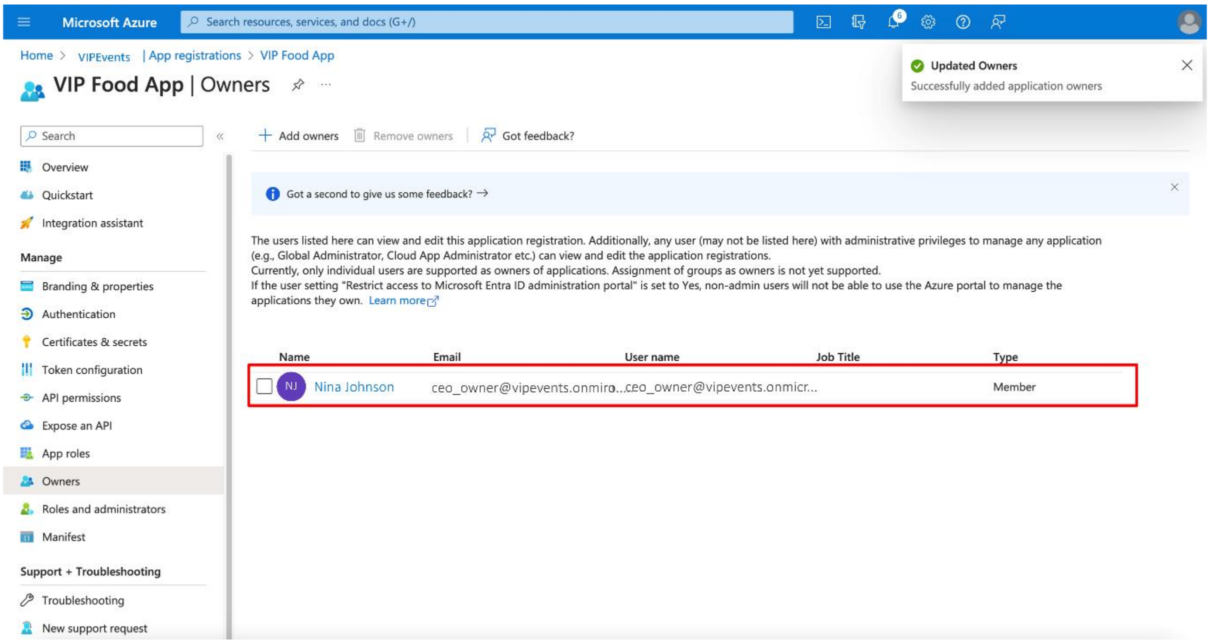
3. Under **Manage**, choose **App roles** and then **Create app role**.
4. Configure role settings and parameters in the **Create app role** pane.



5. Assign Application owners and confirm the selected owner(s).







The values associated with each role, are described below:

- Equipment handlers role:

Role Name: EquipHandlersRole

Details: Designed for equipment handlers, providing access to equipment management functionalities on the VIP Food app.

- Equipment manager role:

Role Name: EquipManagerRole

Details: Tailored for the equipment manager, granting access to overseeing and managing equipment functionalities.

- Chefsrole:

Role Name: ChefsRole

Details: Created for chefs, offering access to food preparation and kitchen management functionalities on the VIP Food app.

- Head chef role:

Role Name: HeadChefRole

Details: Specifically designed for the head chef, providing elevated access to overall kitchen management functionalities.

- Catering manager role:

Role Name: CateringManagerRole

Details: For catering managers, offering access to planning and managing catering events functionalities on the VIP Food app.

- Office workers role:

Role Name: OfficeWorkersRole

Details: Tailored for office workers, providing access to office-related activities and data management functionalities on the VIP Food app.

- CEO (Owner) role:

Role Name: CEORole

Details: Specifically designed for the CEO (owner), granting access to all functionalities on the VIP Food App for strategic oversight.

- Temporary role:

Role Name: TempRole

Details: This role will be assigned to temporary workers, and they will only have access to permission that the user needs for that day – it will be automatically disabled after a period.

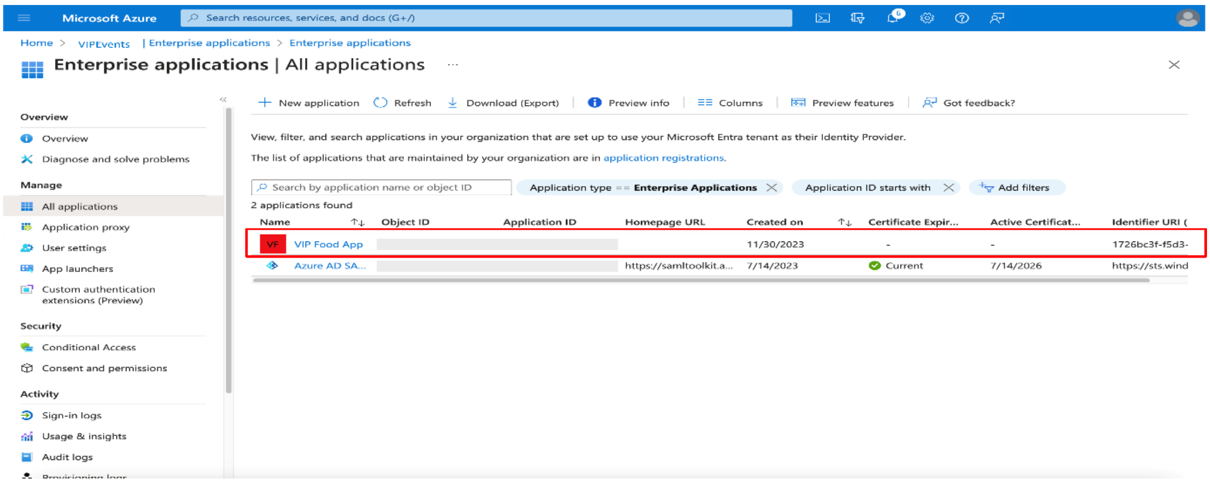
Assigning permissions to Azure roles(Task 3)

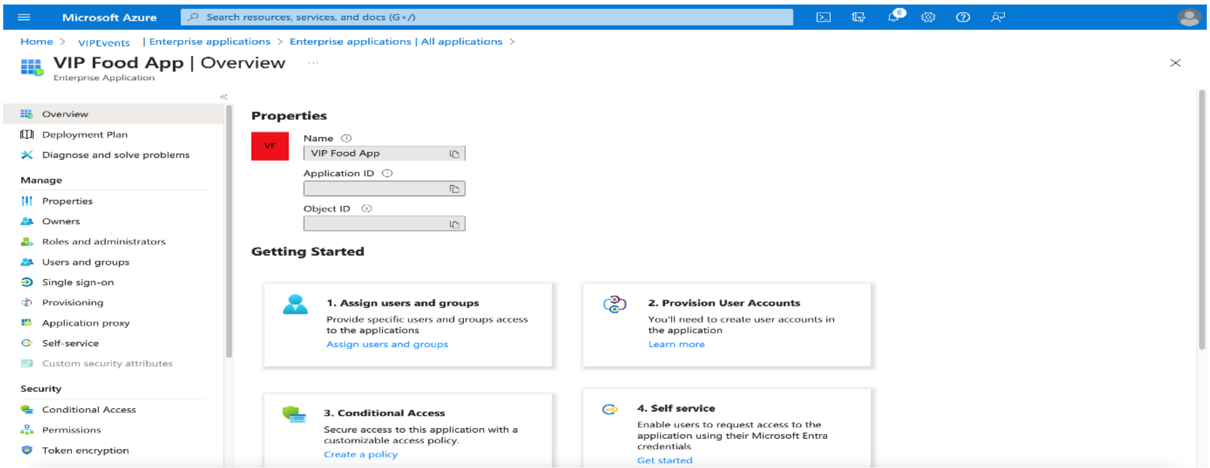
The details of the application and roles should be provided to application developers for the implementation of role-based access control. This step ensures seamless integration of roles into the application structure. They will assign the right role to the right controllers to assign the required permissions.

Mapping user groups to Azure roles (Task 4)

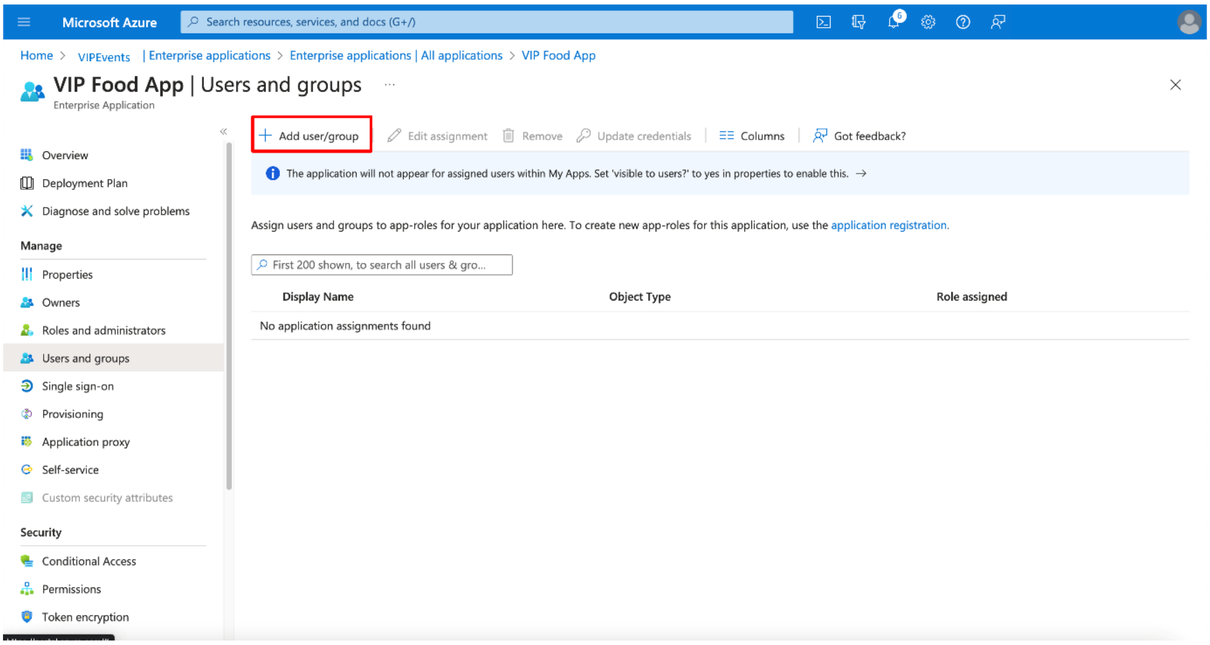
User groups should be mapped to Azure roles to ensure access permissions are tailored to specific roles and responsibilities within the organization. The steps to map user groups to Azure roles are outlined below.

1. Sign in to the **Microsoft Entra admin center**.
2. Go to **Manage**, select **Enterprise Applications** and then select the app.



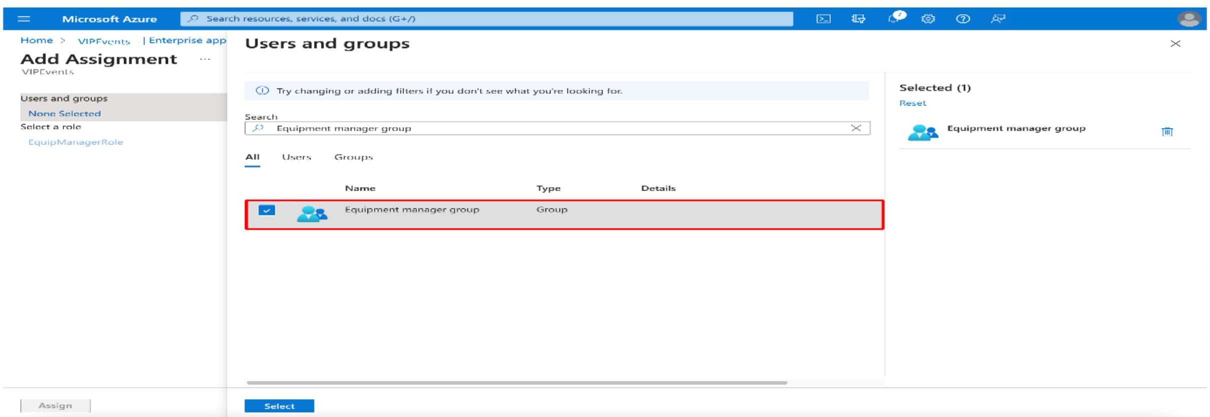


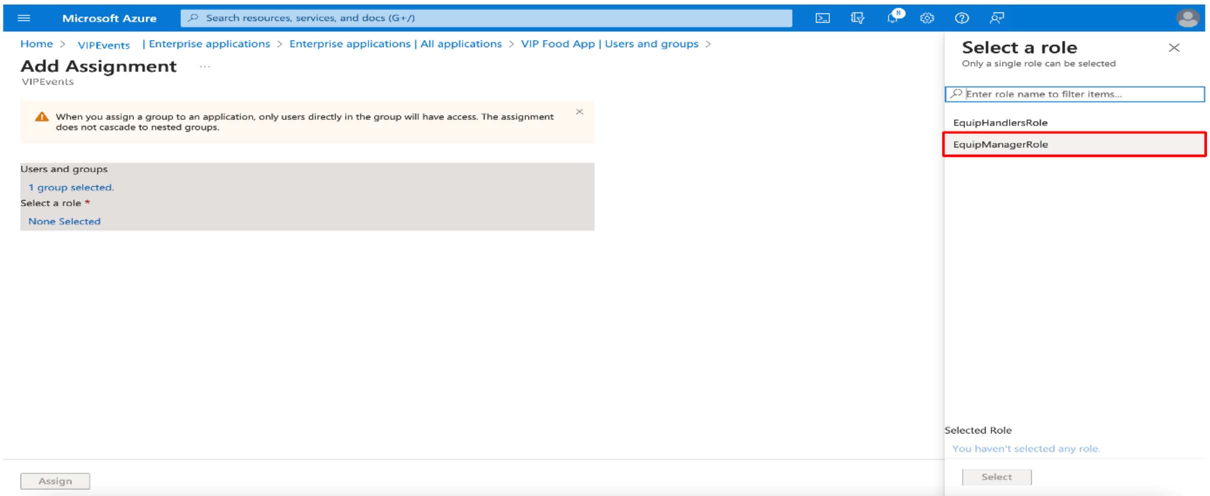
3. Add user group

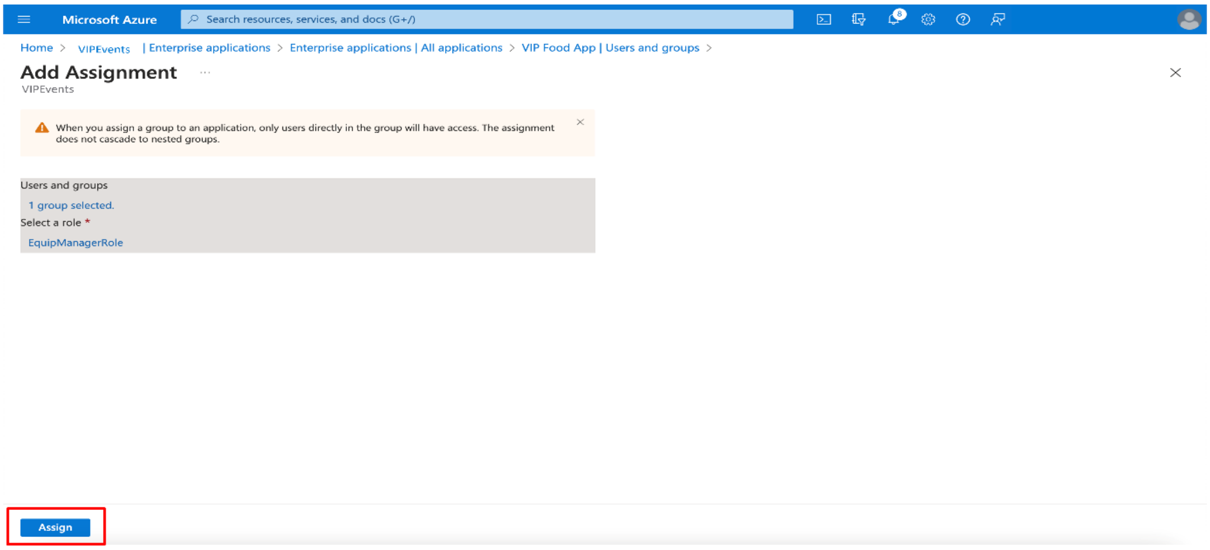


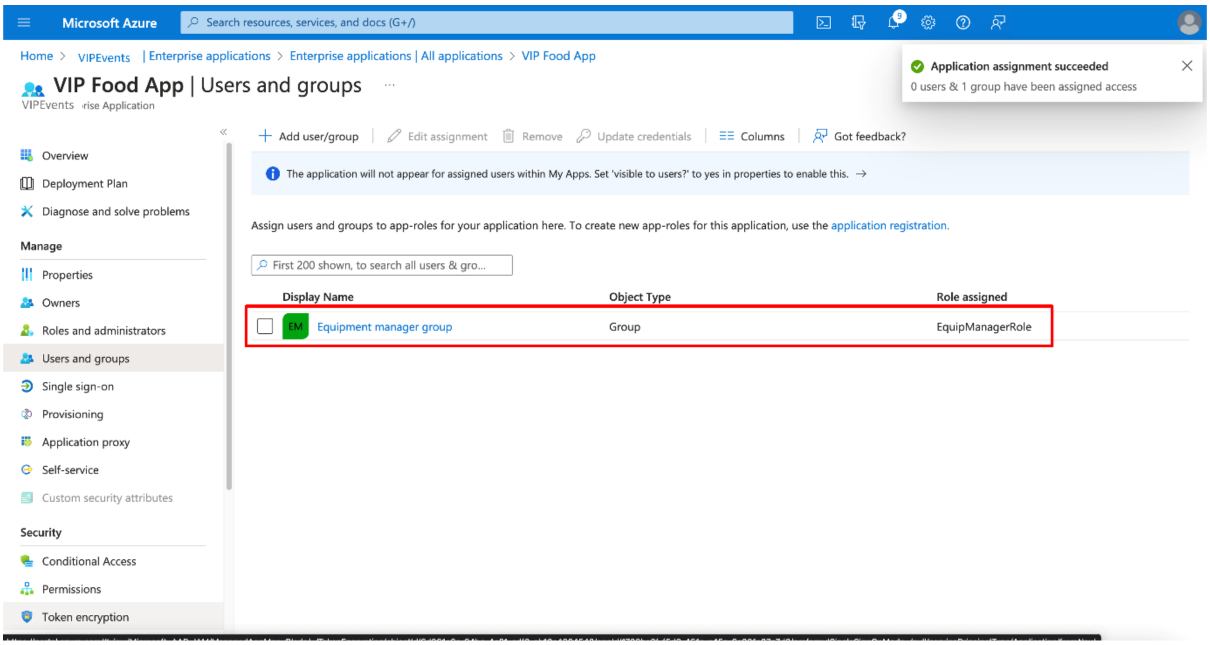
4. Select **Add assignments**.

5. Select the group you want to assign to this role. Use the Role Mapper table provided after the screenshots to know which groups to add to which roles. Note that only role-assignable groups are included in the table. Select the role.




Microsoft Azure
Search resources, services, and docs (G+/)
Home > VIPEvents | Enterprise applications > Enterprise applications | All applications > VIP Food App | Users and groups >
Add Assignment
VIPEvents
When you assign a group to an application, only users directly in the group will have access. The assignment does not cascade to nested groups.
Users and groups
1 group selected.
Select a role *
None Selected
Assign
Select a role
Only a single role can be selected
Enter role name to filter items...
EquipHandlersRole
EquipManagerRole
Selected Role
You haven't selected any role.
Select


Microsoft Azure
Search resources, services, and docs (G+/)
Home > VIPEvents | Enterprise applications > Enterprise applications | All applications > VIP Food App | Users and groups >
Add Assignment
VIPEvents
When you assign a group to an application, only users directly in the group will have access. The assignment does not cascade to nested groups.
Users and groups
1 group selected.
Select a role *
EquipManagerRole
Assign


Microsoft Azure
Search resources, services, and docs (G+/)
Home > VIPEvents | Enterprise applications > Enterprise applications | All applications > VIP Food App
VIP Food App | Users and groups
VIPEvents
Application assignment succeeded
0 users & 1 group have been assigned access
Overview
Deployment Plan
Diagnose and solve problems
Manage
Properties
Owners
Roles and administrators
Users and groups
Single sign-on
Provisioning
Application proxy
Self-service
Custom security attributes
Security
Conditional Access
Permissions
Token encryption
Add user/group
Edit assignment
Remove
Update credentials
Columns
Got feedback?
The application will not appear for assigned users within My Apps. Set 'visible to users?' to yes in properties to enable this.
Assign users and groups to app-roles for your application here. To create new app-roles for this application, use the application registration.
First 200 shown, to search all users & gro...
Display Name
Object Type
Role assigned
Equipment manager group
Group
EquipManagerRole

6. If a group isn't listed, you will need to create a role-assignable group.
7. Select **Add** to assign the role to the group.

Role Mapper table:

Role	**Group to be added**
Equipment handler	Equipment handlersgroup
Equipment manager	Equipment manager group
Chef	Chefs group
Head chef	Head chef group
Catering manager	Catering manager group
Office worker	Office workers group
CEO	CEO (Owner) group
Tempworker (role name)	Worker - external

STAGE 04

AAD Connections (Stage 4)

This section provides specifications for testing Azure AD connections. By performing these tests, VIP Events can verify if new authentication and access procedures are implemented correctly.

Client app integration (Task 1)

- To test whether the Azure login functionality is integrated with the VIP Food app, log into the VIP Food app using a designated test user account.
- To test RBAC implementation, log into the VIP Food app with a user account that has been assigned the relevant Azure role through group membership.
- To verify if the Azure AD Authentication functionality is working, follow the Azure AD authentication prompts.
- Perform functional testing by entering Azure AD credentials and rigorously testing various functionalities of the app to ensure flawless operation.

User account validation (Task 2):

- Begin the Azure user account testing process by logging into the Azure AD portal using the designated credentials.
- Navigate to **User Overview** by going to **Users** and selecting **All users** for a comprehensive overview of user accounts.
- Scrutinize the user list to confirm the creation of all necessary accounts.
- To verify if the user attributes are correct, select a user account, choose **Edit**, and meticulously verify attributes like job title and department.

Application role assignment (Task 3)

- Navigate to **Enterprise Applications** and select **All applications** for an overview.
- Choose the **VIP Food App** and select **Roles and administrators**.
- Review the user-role assignments to ensure alignment with organizational needs.

Group/Role-based access testing (Task 4)

- To start the group-based access testing, create a test user account.
- Assign the test user to a specific user group.
- Test if the role is correctly assigned to the group and if the members of the respective group have the required role assigned automatically. To do this, attempt to access app features applicable to that particular group/role.

MFA verification (Task 5)

- Start by logging into the Azure AD portal.
- To navigate to MFA Settings, go to **Security** and select **Multi-Factor Authentication** for an overview.
- To review MFA configuration, scrutinize the user list to ensure MFA is enabled, and manage user settings for detailed MFA method verification.

Logging and monitoring (Task 7)

- To access Audit Logs navigate to **Monitoring** and select **Audit logs for comprehensive monitoring.**
- Scrutinize audit logs to identify irregular activities such as failed sign-ins or changes to user permissions.

Documentation review (Task 8)

- Meticulously review Azure AD configuration documentation.
- Verify that the setup in the Azure AD portal aligns with the documentation.
- Promptly update documentation to reflect any identified discrepancies.

Policy implementation (Stage 5)

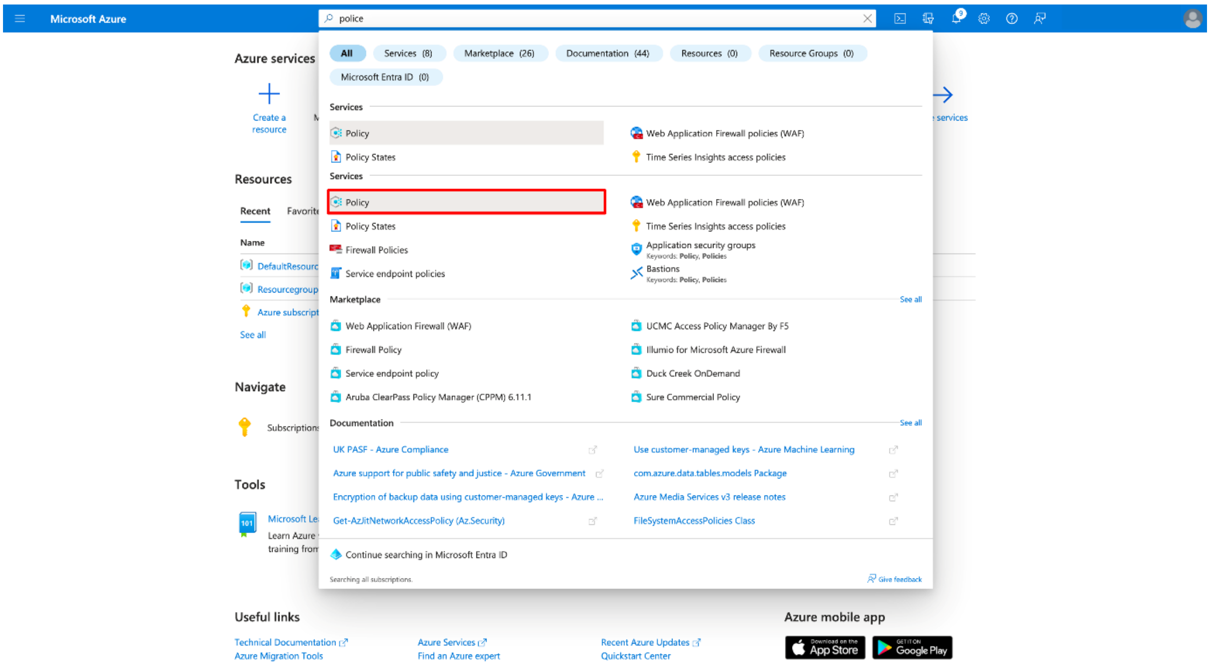
Through the creation and implementation of Azure Policies a robust security framework for VIP Events can be established. Specific policies for user authentication and network configuration for web applications need to be crafted. This section of the proposal provides step-by-step instructions for designing, implementing, and testing these policies to ensure a secure and compliant environment.

User authentication policy (Task 1)

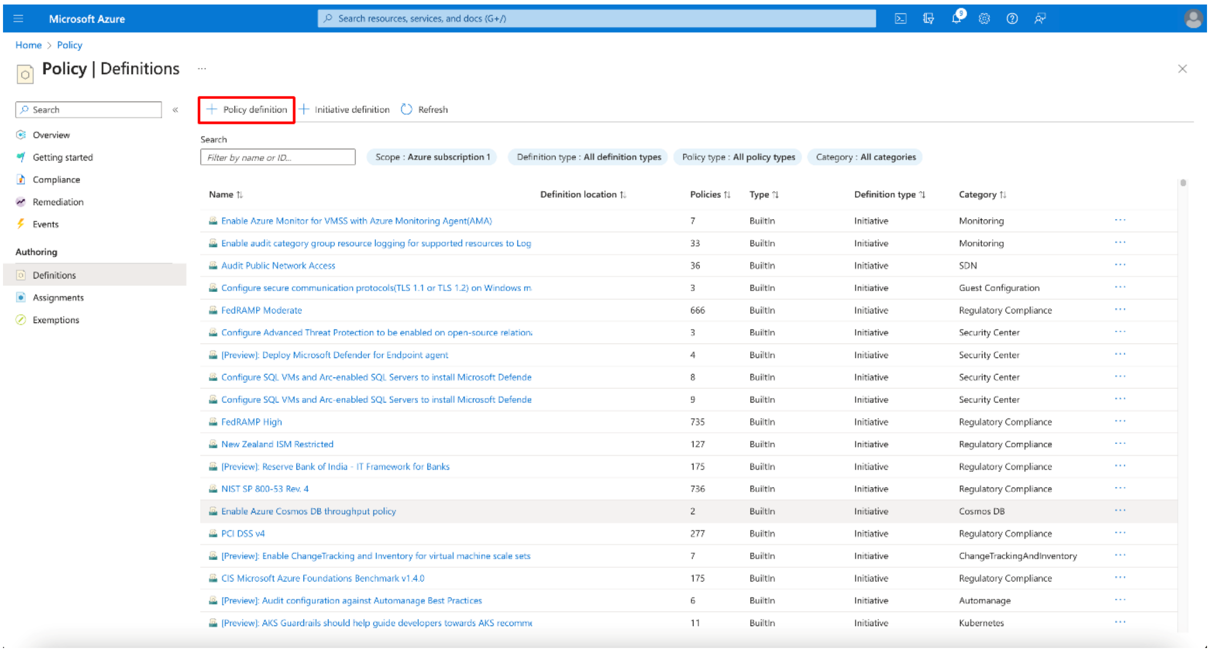
These steps should be followed to create a user authentication policy.

1. Navigate to **Azure Policy** in the Azure Portal.

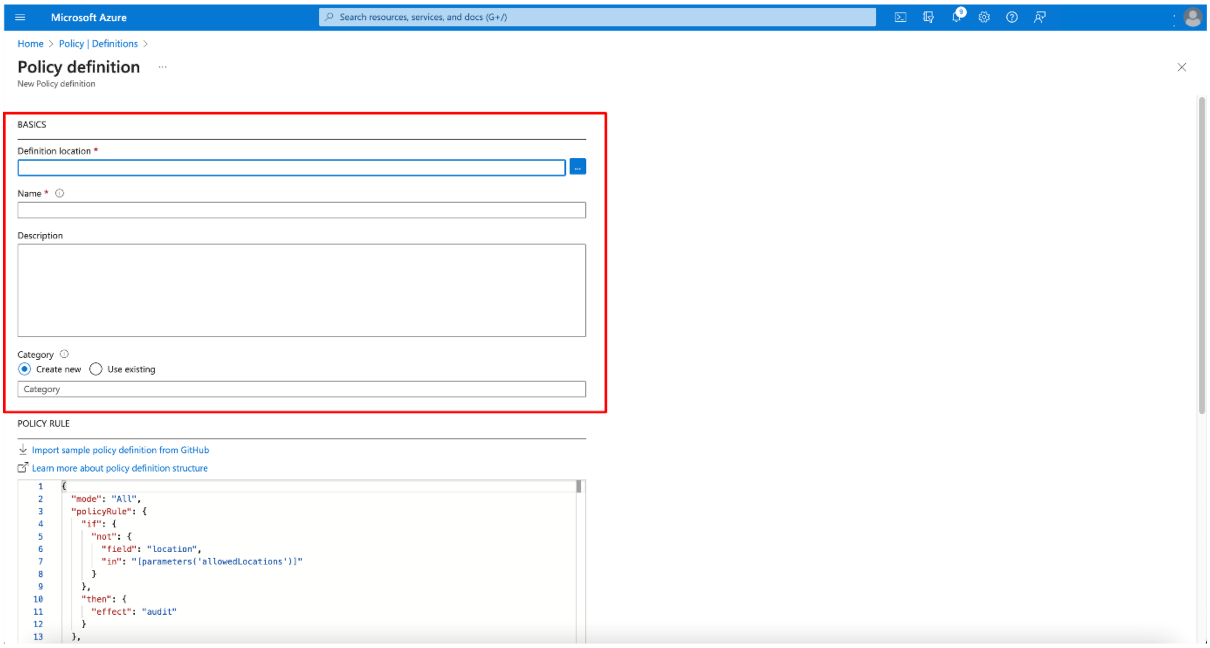
2. In the left-hand menu, select **All services**, and then search for and choose **Policy**.



3. To create a new policy definition, access the **Definitions** tab on the **Policy** page and select **+Policy definition**.



4. Fill in the necessary information like **Display Name**, **Description**, and **Category**.



5. In the **Policy Rule** section, set parameters to enforce MFA for users.
6. Use the **Add a clause** option to specify conditions for the policy, such as requiring MFA for specific user roles or all users.
7. Review the policy settings and select **Review + create**.
8. Ensure validation passes, then select **Create** to create the policy definition.
9. Go to the **Assignments** tab and click on **+Assign policy** to assign the policy to a specific scope. Select the VIP Events subscription.

10. To specify assignment details provide details such as **Scope**, **Parameters**, and **Assignment name**.

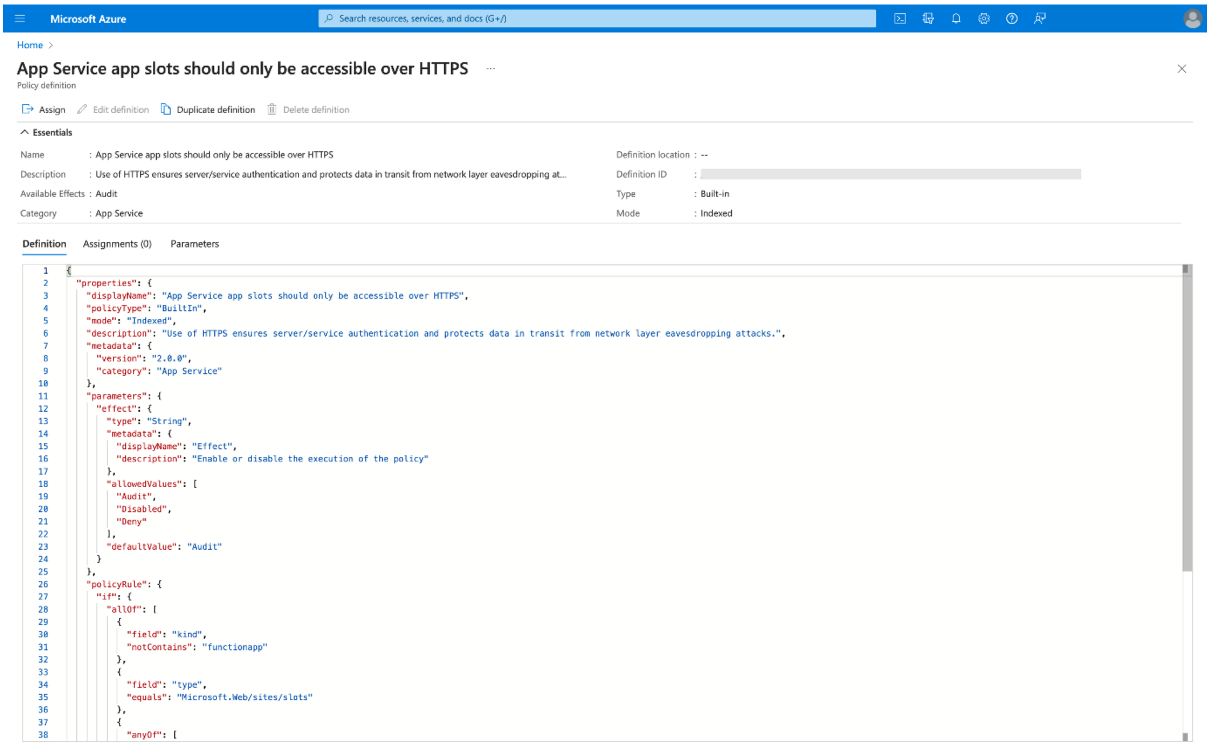
11. Select **Review + assign** and then **Assign** to apply the policy.

Network configuration policy for web applications (Task 2)

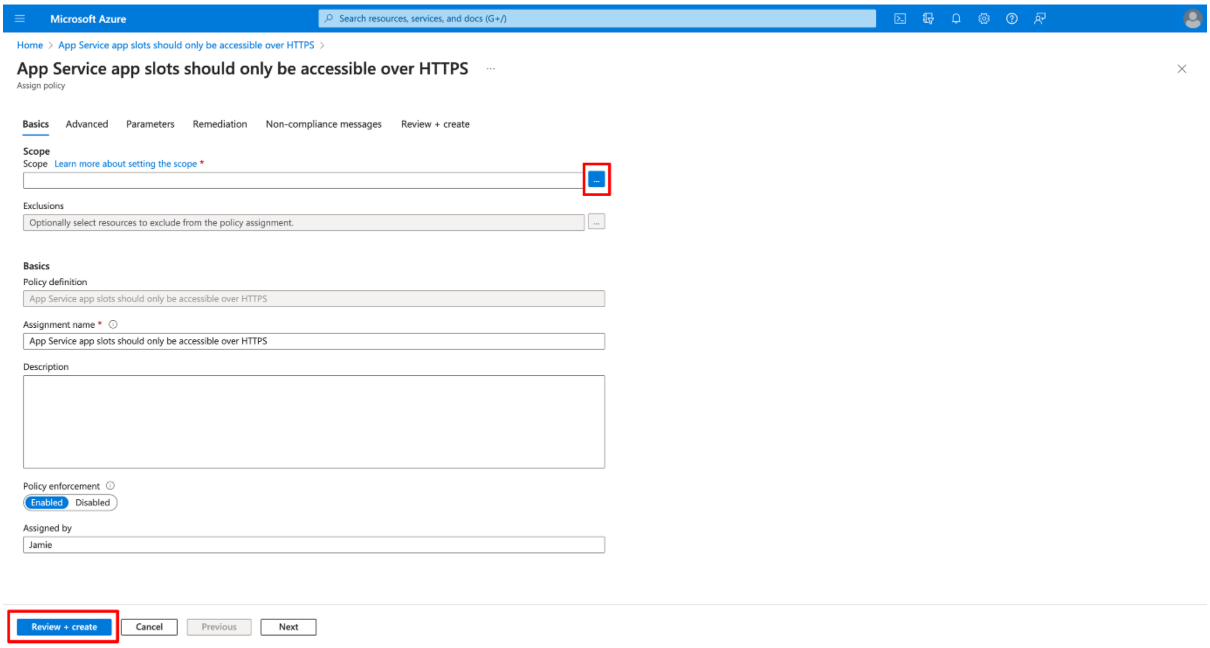
These steps should be followed to configure network configurations for the VIP Food app.

1. Use the built-in policy named **App Service app slots should only be accessible over HTTPS**.

2. This policy can be accessed via this link.



3. Select **Assign**, then choose **Scope** and opt for VIP Food App Subscription.



1. Click Select **Review+Create** to assign the same

Testing in a non-production environment (Task 3)

Before applying policies to production resources, testing in a non-production environment is crucial. Testing identifies unintended consequences or disruptions, verifies policy effectiveness, and ensures alignment with organizational requirements.

Implementing policies within a system carries inherent risks that can compromise its functionality and security. There is a possibility that policies may inadvertently block access for legitimate users or resources, leading to disruptions in normal operations. Unforeseen issues in policy configuration may arise, causing service disruptions and potentially compromising the integrity of the system. Testing policies directly in a production environment can pose security and compliance risks.

The benefits of testing policies before full implementation are substantial. Early detection of misconfigurations or conflicts in policies allows for timely adjustments, ensuring a smoother implementation process. Testing also verifies the impact of policies on resources and user access, helping to identify potential issues before they escalate. By testing policies before implementation, organizations can minimize disruptions to production services, promoting a more stable and secure operational environment.

Through comprehensive policies and thorough testing in a non-production environment, the organization can mitigate security risks, protect sensitive data, and ensure the integrity of its IT systems and operations. Regular monitoring and updates to policies will further enhance the security posture over time.

Conclusion

This is a summary of the cybersecurity proposal for VIP Events:

In the company requirements section of the proposal, crucial information about VIP Events' business operations, security requirements, and user roles was summarized, laying the groundwork for the subsequent stages. The proposal included the creation of an Azure AD tenant for VIP Events. It also detailed how user accounts were to be configured and group-based access control implemented, in line with the specific security requirements identified.

Further, the proposal outlined roles and access configuration. It specified how access permissions for each user group should be finely adjusted to ensure precision, guaranteeing that users have the appropriate level of access for the seamless execution of their job responsibilities.

The proposal also covered the integration of Azure AD with key business applications, particularly the VIP Food app. This integration was designed to streamline user authentication and authorization, fostering a cohesive and efficient operational environment.

Lastly, the proposal recommended implementing certain policy configurations to help VIP Events mitigate security risks, protect sensitive data, and ensure the integrity of its IT systems and operations. The policies described are intended to govern user behavior, regulate data access, and manage devices, thereby forming a robust framework to safeguard VIP Events' cybersecurity landscape.

In summary, the proposal outlines a structured and comprehensive approach to strengthen VIP Events' security framework, safeguarding against potential threats and ensuring ongoing compliance with industry standards.